2. Operators

1. Write a function for arithmetic operators(+,-,*,/).
Answer:
```
public class ArithmeticOperators {

    public static void arithmetic(int a, int b) {
        System.out.println("Addition: " + (a + b));
        System.out.println("Subtraction: " + (a - b));
        System.out.println("Multiplication: " + (a * b));
        System.out.println("Division: " + (a / b));
    }

    public static void main(String[] args) {
        arithmetic(20, 10);
    }
}
```

2. Write a method for increment and decrement operators(++, --).
Answer:
```
public class IncrementDecrement {

    public static void incrementDecrement(int num) {
        System.out.println("Original Number: " + num);

        num++;
        System.out.println("After Increment: " + num);

        num--;
        System.out.println("After Decrement: " + num);
    }

    public static void main(String[] args) {
        incrementDecrement(10);
    }
}
```

3. Program to equal operator and not equal operators.
Answer:
```
public class EqualNotEqual {

    public static void main(String[] args) {
        int a = 10;
        int b = 20;

        System.out.println("a == b : " + (a == b));
        System.out.println("a != b : " + (a != b));
    }
}
```

4. Write a program to find the two numbers **equal or not**.

Answer:

```
public class CheckEqual {

    public static void main(String[] args) {
        int num1 = 25;
        int num2 = 25;

        if (num1 == num2) {
            System.out.println("Both numbers are equal");
        } else {
            System.out.println("Both numbers are not equal");
        }
    }
}
```

5. Programs on Logical AND,OR operator and Logical NOT.

Answer:

```
public class LogicalOperators {

    public static void main(String[] args) {
        boolean a = true;
        boolean b = false;

        // Logical AND
        System.out.println("a && b : " + (a && b));

        // Logical OR
        System.out.println("a || b : " + (a || b));

        // Logical NOT
        System.out.println("!a : " + (!a));
        System.out.println("!b : " + (!b));
    }
}
```

6. Program for relational operators (<,<==, >, >==).

Answer:

```
public class RelationalOperators {

    public static void main(String[] args) {
        int a = 15;
        int b = 10;

        System.out.println("a < b : " + (a < b));
        System.out.println("a <= b : " + (a <= b));
        System.out.println("a > b : " + (a > b));
        System.out.println("a >= b : " + (a >= b));
    }
}
```

7. Print the smaller and larger number

Answer:

```
public class SmallLargeNumber {

    public static void main(String[] args) {
        int a = 30;
        int b = 50;

        if (a > b) {
            System.out.println("Larger Number: " + a);
            System.out.println("Smaller Number: " + b);
        } else {
            System.out.println("Larger Number: " + b);
            System.out.println("Smaller Number: " + a);
        }
    }
}
```